RELATIONAL MODEL

Database Management System (DBMS) and Programming

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A Brief of Relational Models

- Introduced by Ted Codd of IBM Research in 1970 in a classic paper (Codd,1970)
- Based on the concept of a mathematical relation
- Commercial implementations were available in the early 1980s, such as the SQL/DS system on the MVS operating system by IBM and the Oracle DBMS, eventually open source community by MySQL, MariaDB, or ProsgreSQL

Relational Data Structure

- Relation
- Attribute
- Domain
- Tuple
- Degree
- Cardinality
- Relational database

Relational Data Structure

1. Relation

- A relation is a table with columns and rows.

Relation name

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Relational Data Structure

2. Attribute

- An attribute is a named column of a relation.

Attributes

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Relational Data Structure

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

$$R(A_1, A_2, \ldots, A_n)$$

- R is a name of relation
- $A_1, A_2, \ldots, A_n$ are the name of the attribute
- EX.

$$Employee(EmpID, Firstname, Lastname, Salary)$$

Relational Data Structure

3. Domain

- A domain is the set of allowable values for one or more attributes.
- Every attribute in the relation is defined on a domain.

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Attribute	Meaning	Domain Definition
EmpID	The set of all possible employee numbers	Integer: size 10, range 1-9999999999
Fitstname	The set of all possible first name of employee	Character: size 50
Lastname	The set of all possible last name of employee	Character: size 50
Salary	Possible value of employee salaries	Float: 7 digits, range 1000.00 - 9999999.99

Relational Data Structure

4. Tuple

- A tuple is a row of a relation.

  $\mathtt{Tuple} = \langle v_1, v_2, \ldots, v_n \rangle$

- $v_1, v_2, \ldots, v_n$ are the value of each corresponding attributes
- Ex.

```
<1, ”John”, “Smith”, 30000>
<2, ”Franklin”, “Wong”, 40000>
```

Tuples

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Relational Data Structure

5. Degree

- The degree of a relation is the number of attributes it contains

6. Cardinality

- The cardinality of a relation is the number of tuples it contains

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Cardinality

Degree

Relational Data Structure

7. Relational database

- A collection of normalized relations with distinct relation names.

Relation Schema

- A named relation defined by a set of attribute and domain name pairs.

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

```
{(John, Smith, 30000)}
{(Firstname: John, Lastname: Smith, Salary: 30000)}
```

Mathematical Relations

- Given 2 sets
  - $D_1 = \{1, 2\}$
  - $D_2 = \{A, B\}$

- The Cartesian product of these 2 sets
  - $D_1 \times D_2 = \{(1,A), (1,B), (2,A), (2,B)\}$

Mathematical Relations

- Any subset of the Cartesian product is a relation.

  $R \subseteq D_1 \times D_2$

  $R = \{(1,A), (2,A)\}$

  $R = \{(1,A), (2,A), (2,B)\}$

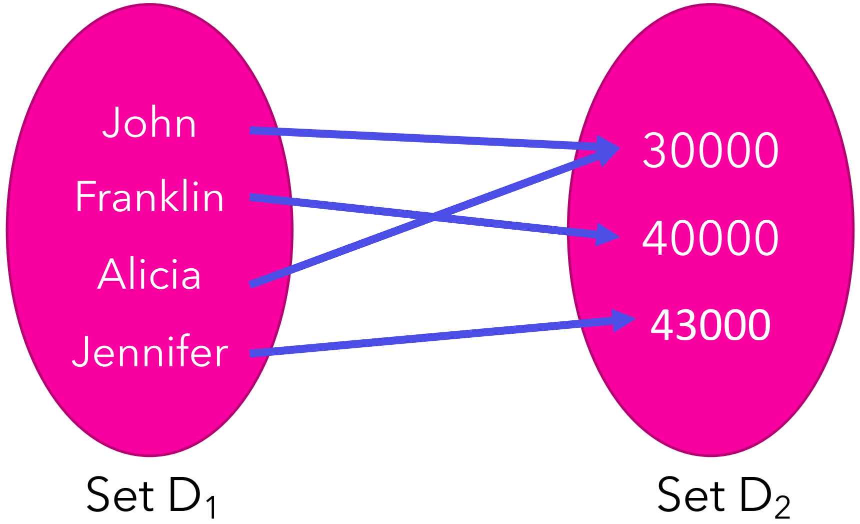
- Or, we can specify the member of the relation by setting some conditions.

  $R = \{x,y \mid x \in D_1 \wedge y \in D_2\}$

- It can be extended beyond 2 domains.

  $R \subseteq D_1 \times D_2 \times \cdots \times D_n$

Firstname	Salary
John	30000
Franklin	40000
Alicia	30000
Jennifer	43000



R = {(John,30000), (Franklin, 40000) , (Alicia, 30000) , (Jennifer, 43000)}

Alternative Terms

Formal Terms (Codd)	Alternative Term 1	Alternative Term 2
Relation	Table	File
Tuple	Row	Record
Attribute	Column	Field

Properties of Relations

Properties of Relations

- The *order* of tuples has *no significance*.
  - A Relation is a set of tuples.
  - Tuples in relation does not have any particular order.

Employee

ID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

A = {1,2,3,4}

SET A = B

Employee

ID	Firstname	Lastname	Salary
3	Alicia	Zelaya	29000
1	John	Smith	30000
4	Jennifer	Wallace	43000
2	Franklin	Wong	40000

B = {3,1,4,2}

Relational data structure

- The order of attributes in a relation has no significance

Employee

ID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Employee

ID	Lastname	Salary	Firstname
1	Smith	30000	John
2	Wong	40000	Franklin
3	Zelaya	29000	Alicia
4	Wallace	43000	Jennifer

`A = {(Firstname: `**`John`**`, Lastname: `**`Smith`**`, Salary: `**`30000`**`)}`

`B = {(Lastname: `**`Smith`**`, Salary: `**`30000`**`, Firstname: `**`John`**`)}`

SET A = B

Properties of Relations

- The relation has a name that is distinct from all other relation names.

Employee

EmpID	Firstname	Lastname	Salary	DeptID
1	John	Smith	30000	1
2	Franklin	Wong	40000	1
3	Alicia	Zelaya	29000	2
4	Jennifer	Wallace	43000	3

Department

DeptID	DeptName
1	Sale
2	HR
3	Research

Properties of Relations

- Each cell of the relation contains exactly one ***atomic*** value.

Employee

EmpID	Firstname	Lastname	Salary	Address	Dept_ID
1	John	Smith	30000	Mueang, Chiang Mai 50100	1
2	Franklin	Wong	40000	Khlong Toei, Bangkok 10100	1
3	Alicia	Zelaya	29000	Bang Rak, Bangkok 10200	2
4	Jennifer	Wallace	43000	Mueang, Chiang Mai 50000	3

EmpID	Firstname	Lastname	Salary	City	Province	Postcode	Dept_ID
1	John	Smith	30000	Mueang	Chiang Mai	50100	1
2	Franklin	Wong	40000	Khlong Toei	Bangkok	10100	1
3	Alicia	Zelaya	29000	Bang Rak	Bangkok	10200	2
4	Jennifer	Wallace	43000	Mueang	Chiang Mai	50000	3

Properties of Relations

- Nulls value
  - Unknown OR Not applicable (value exist but not available)

No	Firstname	Lastname	Salary	Address	Dept_ID
1	John	Smith	30000	Mueang, Chiang Mai 50100	1
2	Franklin	***null***	40000	Khlong Toei, Bangkok 10100	1
3	Alicia	Zelaya	29000	***null***	2
4	Jennifer	Wallace	43000	Mueang, Chiang Mai 50000	3

Not applicable

Unknown

Properties of Relations

- Each attribute has distinct name.

Attributes

Employee

EmpID	Firstname	Lastname	Salary	City	Province	Postcode	Dept_ID
1	John	Smith	30000	Mueang	Chiang Mai	50100	1
2	Franklin	Wong	40000	Khlong Toei	Bangkok	10100	1
3	Alicia	Zelaya	29000	Bang Rak	Bangkok	10200	2
4	Jennifer	Wallace	43000	Mueang	Chiang Mai	50000	3

Properties of Relations

- The values of an attribute are all from the same domain.

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

Attribute	Meaning	Domain Definition
EmpID	The set of all possible employee numbers	Integer: size 10, range 1-9999999999
Fitstname	The set of all possible first name of employee	Character: size 50
Lastname	The set of all possible last name of employee	Character: size 50
Salary	Possible value of employee salaries	Float: 7 digits, range 1000.00 - 9999999.99

Properties of Relations

- Each tuple is ***distinct***; there are no duplicate tuples.

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
4	Jennifer	Wallace	43000

$A = \{1,2,3,4\}$

Employee

EmpID	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
3	Alicia	Zelaya	29000

$A = \{1,2,3,3\}$

RELATIONAL KEY

Relational Key

- Relational key is one or more attributes that uniquely identifies each tuple in a relation.
  - Super key
  - Candidate key
  - Primary key
  - Foreign key

Super key

- A set of attributes within a table that can *uniquely identify* each record within a table.
- Every relation has at least one super key

Employee

EmpID	Firstname	Phome	Age
1	John	9876723452	30
2	Franklin	9991165674	30
3	Alicia	7898756543	50
4	John	8987867898	25

SK = EmpID, (EmpID, Firstname), phone

Super key

Employee

EmpID	Firstname	Lastname	Email
1	John	Smith	aaa@aa.com
2	Franklin	Wong	bbb@bb.com
3	Alicia	Zelaya	ccc@cc.com
4	Jennifer	Wallace	ddd@dd.com

What are the super key of this relation?

```
SK = {EmpID}, {EmpID, Firstname},  {EmpID, Lastname},
{EmpID, Firstname, Lastname},
{Email}, {EmpID, Email}, {Email, Firstname},  {Email, Lastname},
{Email, Firstname, Lastname},
{EmpID, Email, Firstname}, {EmpID, Email, Lastname}
{EmpID, Firstname, Lastname, Email, Lastname}
```

Candidate Key

- A Candidate key is a subset of *Super keys*
- The *minimal set* of fields which can uniquely identify each record in a table
- It is an attribute or a set of attributes that can act as a *Primary Key*
- There can be more than one candidate key.

Employee

EmpID	Firstname	Lastname	Email
1	John	Smith	aaa@aa.com
2	Franklin	Wong	bbb@bb.com
3	Alicia	Zelaya	ccc@cc.com
4	Jennifer	Wallace	ddd@dd.com

CK = {EmpID}, {Email}

```
SK = {EmpID}, {EmpID, Firstname},  {EmpID, Lastname},
{EmpID, Firstname, Lastname},
{Email}, {EmpID, Email}, {Email, Firstname},  {Email, Lastname},
{Email, Firstname, Lastname},
{EmpID, Email, Firstname}, {EmpID, Email, Lastname}
{EmpID, Firstname, Lastname, Email, Lastname}
```

Primary Key

- The candidate key that is selected to identify tuples uniquely with in the relation

Employee

EmpID	Firstname	Lastname	Email
1	John	Smith	aaa@aa.com
2	Franklin	Wong	bbb@bb.com
3	Alicia	Zelaya	ccc@cc.com
4	Jennifer	Wallace	ddd@dd.com

CK = {EmpID}, {Email}

PK = {EmpID}

Primary Key

- In the relation, the key can be indicated by underlining the attribute.

Employee

<u>EmpID</u>	Firstname	Lastname	Email
1	John	Smith	aaa@aa.com
2	Franklin	Wong	bbb@bb.com
3	Alicia	Zelaya	ccc@cc.com
4	Jennifer	Wallace	ddd@dd.com

`Employee(EmpID, Firstname, Lastname, Email)`

Primary key

Unique

Foreign Key

- An attribute, or set of attributes, within one relation that match the candidate key of some relation.

Foreign Key

Employee

<u>EmpID</u>	Firstname	Lastname	Salary	DeptID
1	John	Smith	30000	1
2	Franklin	Wong	40000	1
3	Alicia	Zelaya	30000	2
4	John	Wallace	43000	3

Department

<u>DeptID</u>	DeptName
1	Sale
2	HR
3	Research

FK = { DeptID }

Foreign Key

- Foreign key (FK) must satisfy the following:
  - Same domain
  - Value of foreign key in a tuple either occurs as value of primary key (PK)
  - $table_1[FK] = table_2[PK]$ or is null

Foreign Key

Employee

EmpID	Firstname	Lastname	Salary	DeptID
1	John	Smith	30000	1
2	Franklin	Wong	40000	1
3	Alicia	Zelaya	30000	2
4	John	Wallace	43000	3

Department

DeptID	DeptName
1	Sale
2	HR
3	Research

Composite Key

- Composite key is a candidate key that consists of two or more attributes that together uniquely identify an entity occurrence.
- The Primary Key which defined using more that one column.

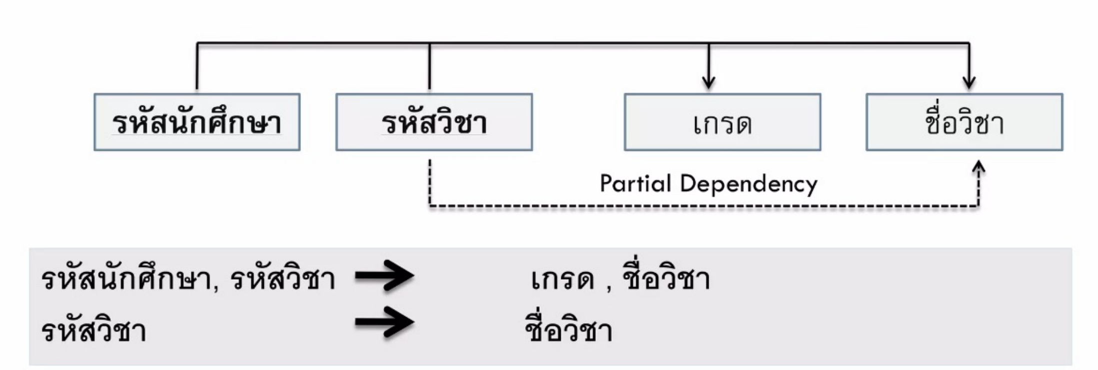
Employee

EmpID	Firstname	Lastname	Salary	DeptID
1	John	Smith	30000	1
2	Franklin	Wong	40000	1
3	Alicia	Zelaya	30000	2
4	John	Wallace	43000	3

Composite Key

Partial Dependency

- The primary key has more than 1 attributes
- Part of the primary key can identify the value of other attribute in the relation



Transitive Dependency

- Non-primary key attribute depends on another Non-primary key attribute in the relation

เลขประจำตัว	ชื่อ สกุล	ที่อยู่	ตำแหน่ง	รถประจำตำแหน่ง
01	ฉัตรชัย มีสมบัติ	กรุงเทพ	ผู้จัดการ	BMW
02	เอกชัย ใจดี	นนทบุรี	ผู้ช่วยผู้จัดการ	Honda
03	มนีรัตน์ เจริญสุข	เชียงใหม่	ผู้จัดการ	BMW
04	ขวัญชัย ใจเพชร	ราชบุรี	ผู้ช่วยผู้จัดการ	Honda

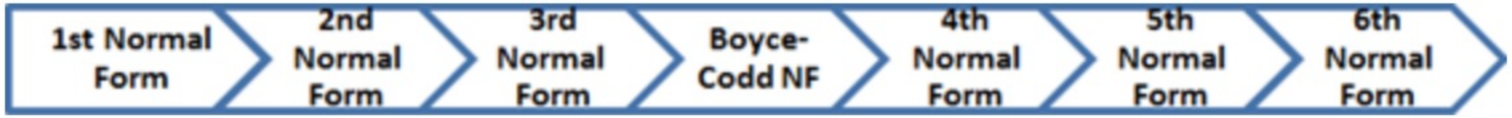
คำอธิบาย เลขประจำตัว เป็นคีย์หลัก (Primary Key) ของตาราง

เลขประจำตัว $\longrightarrow$ ชื่อสกุล, ที่อยู่, ตำแหน่ง

ตำแหน่ง $\longrightarrow$ รถประจำตำแหน่ง

Normalization

- Helps to group related data in one single table
- Normalization increases *data consistency* as it *avoids* data duplicity by storing it in one place only.
- Normalization helps in grouping like or related data under the *same* schema.
- Improves searching faster



Database Normal Forms

Example

Full Names	Physical Address	Movies rented	Salutation
Janet Jones	First Street Plot No 4	Pirates of the Caribbean, Clash of the Titans	Ms.
Robert Phil	3rd Street 34	Forgetting Sarah Marshal, Daddy's Little Girls	Mr.
Robert Phil	5th Avenue	Clash of the Titans	Mr.

EMPLOYEE_ID	NAME	JOB_CODE	JOB	STATE_CODE	HOME_STATE
E001	Alice	J01	Chef	26	Michigan
E001	Alice	J02	Waiter	26	Michigan
E002	Bob	J02	Waiter	56	Wyoming
E002	Bob	J03	Bartender	56	Wyoming
E003	Alice	J01	Chef	56	Wyoming

1 NF

- A single cell must not hold more than one value (atomicity)
- There must be a primary key for the identification
- No duplicated rows or columns
- Each column must have only one value for each row in the table

EMPLOYEE_ID	NAME	JOB_CODE	JOB	STATE_CODE	HOME_STATE
E001	Alice	J01	Chef	26	Michigan
E001	Alice	J02	Waiter	26	Michigan
E002	Bob	J02	Waiter	56	Wyoming
E002	Bob	J03	Bartender	56	Wyoming
E003	Alice	J01	Chef	56	Wyoming

2 NF

- It's already in 1NF
- Has no partial dependency. That is, all non-key attributes are fully dependent on a primary key.

Employee table

EMPLOYEE_ID	NAME	STATE_CODE	HOME_STATE
E001	Alice	26	Michigan
E002	Bob	56	Wyoming
E003	Alice	56	Wyoming

Job table

JOB_CODE	JOB
J01	Chef
J02	Waiter
J03	Bartender

EmpRole table

EMPLOYEE_ID	JOB_CODE
E001	J01
E001	J02
E002	J02
E002	J03
E003	J01

3 NF

Employee table

EMPLOYEE_ID	NAME	STATE_CODE
E001	Alice	26
E002	Bob	56
E003	Alice	56

State table

STATE_CODE	HOME_STATE
26	Michigan
56	Wyoming

Job table

JOB_CODE	JOB
J01	Chef
J02	Waiter
J03	Bartender

EmpRole table

EMPLOYEE_ID	JOB_CODE
E001	J01
E001	J02
E002	J02
E002	J03
E003	J01

3 NF

- Be in 2NF
- Have no transitive partial dependency.

Employee table

EMPLOYEE_ID	NAME	STATE_CODE	SaluationID
E001	Alice	26	2
E002	Bob	56	1
E003	Alice	56	3

State table

STATE_CODE	HOME_STATE
26	Michigan
56	Wyoming

Saluation table

SaluationID	Saluation
1	Mr.
2	Miss
3	Dr.

Job table

JOB_CODE	JOB
J01	Chef
J02	Waiter
J03	Bartender

EmpRole table

EMPLOYEE_ID	JOB_CODE
E001	J01
E001	J02
E002	J02
E002	J03
E003	J01